

Measurement of Terahertz Band Water Vapor Absorption Spectrum for the Realization of High-Precision Optical Humidity Measurement

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1. Research Background

Humidity is a familiar physical quantity, even to elementary school students. Recently, the relationship between temperature and humidity has gained attention in relation to illnesses like viral infections and influenza. Humidity is also considered crucial in various fields, such as managing low-humidity environments in battery manufacturing for electric vehicles as part of the move toward "decarbonization," and its effects on the quality and efficacy of pharmaceuticals due to the transition of crystal polymorphs into hydrates and anhydrides.

Commercially available hygrometers currently have extremely low accuracy. The humidity accuracy of a typical thermo-hygrometer is $\pm 5\%$, and even high-precision models are only around $\pm 2\%$. Furthermore, errors tend to increase when the object's temperature is outside the measurement range or in low-humidity conditions, leading to accuracy worse than stated above. High-precision measurement of trace gases in the atmosphere is performed using non-dispersive infrared absorption (NDIR), an optical measurement method. However, it is not yet practical for water vapor, which has strong absorption in the mid-infrared region.

In recent years, terahertz waves—electromagnetic waves with frequencies from 100 GHz to 10 THz (corresponding to wavelengths from 1 mm to 0.01 mm) located between light and radio waves—have been gaining attention. Terahertz spectroscopy using these waves can detect intermolecular

vibrations and crystal lattice motions (phonon modes) in solids. This allows for the observation of the entire molecule's vibration, unlike in the infrared band where identifying polymorphs of water and alcohol by observing the O-H vibration of water molecules can be insufficient. Furthermore, several independent absorption lines unique to water vapor exist across the broad spectrum of the terahertz band. The goal is to achieve high-precision absolute humidity measurement over a wide dynamic range by selecting multiple of these absorption lines and measuring the transmitted terahertz intensity.

2. Purpose

To realize an optical hygrometer with an accuracy more than an order of magnitude better than existing hygrometers, which have an accuracy of $\pm 5\%$ (or $\pm 2\%$ for high-precision models), by using terahertz spectroscopy to measure and evaluate the water vapor absorption spectrum.

3. Principle

A Michelson interferometer was used for the water vapor spectrum measurement. By measuring the intensity of the transmitted light at various points as the movable mirror moves, an interferogram is obtained. A spectrum can then be derived by applying a Fourier transform to this interferogram.

$I_{sam}(t)$ be the interferogram with the measurement sample present and $I_{ref}(t)$ be the interferogram without the sample (background), the transmittance T and

absorbance A can be expressed by the following equations

Transmittance

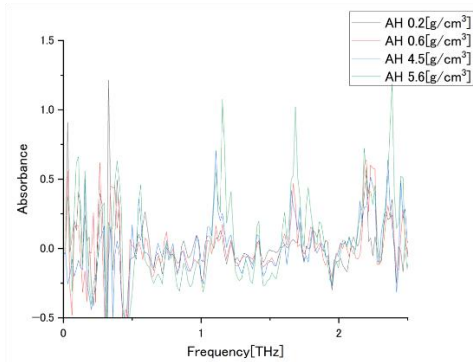
$$T = \frac{\int I_{sam}(t)e^{-j\omega t} dt}{\int I_{ref}(t)e^{-j\omega t} dt} \quad (1)$$

Absorbance

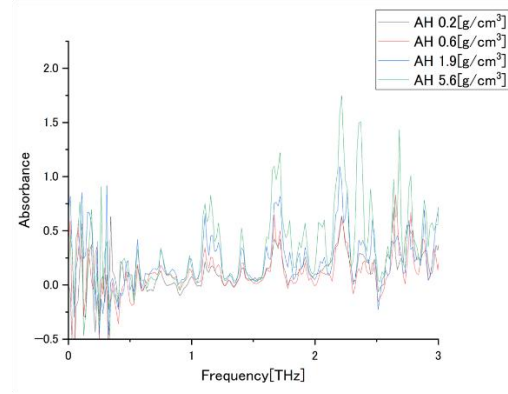
$$A = \log \frac{1}{T} = \log \frac{\int I_{ref}(t)e^{-j\omega t} dt}{\int I_{sam}(t)e^{-j\omega t} dt}$$

4. Results

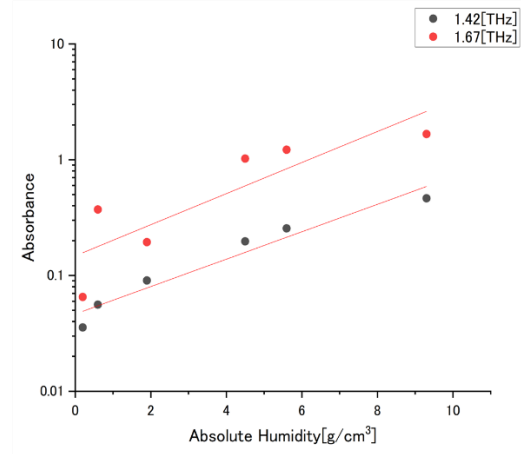
Interferograms were measured for each relative humidity level and for dry air (approx. 0.2%) using a Michelson interferometer. The absorption spectra were then calculated by performing a Fourier transform using OriginPro (Light Stone).



Graph 1: Water vapor absorption spectrum (measured July 26, 2023).



Graph 2: Water vapor absorption spectrum (measured July 6, 2024)



Graph 3: Regression line by least squares method

The measurable range was approximately 0.5 to 2.0 THz. Absorption peaks observed in both measurements were at 0.75, 1.10, 1.15, 1.21, 1.42, and 1.67 THz. These peak values are considered to be the intrinsic absorption spectra of water vapor. Peaks observed only once in either the past or current measurements cannot be definitively identified as intrinsic water vapor peaks. These discrepancies are thought to be due to the quality of the beam (light source) and the

influence of the optical path difference from adjustments to the optical system.

5. Discussion

After calculating the absolute error between the regression line and the measured values, it was found that among the six absorption peaks identified from the measurements, the one at 1.42 THz could be measured with a standard deviation of about 0.04. This means that regarding the objective of precision, there is an error of about 4% from the regression line derived by the least squares method. While a high precision of about 0.1% was achieved at an absolute humidity of 0.6 g/cm³, an error of 4% or more occurred at absolute humidities of 4.5 g/cm³ and higher. It is thought that increasing the number of humidity points for measurement could improve the accuracy of the regression line, thereby reducing the error. Through this measurement, we believe that the absorbance at 1.42 THz is crucial for the realization of a high-precision optical humidity meter.

研究背景

<日本語文>

また、「脱炭素」に向けた電気自動車のバッテリー製造における低湿度環境の管理や、医薬品の結晶多形が水和物・無水物へ転移することによる品質や有効性への影響など、様々な分野で重要視されている。

DeepL での翻訳

It is also important in a variety of fields, such as the management of low-humidity environments in the manufacture of batteries for electric vehicles for decarbonization, and the impact on quality and efficacy of the transition of crystalline polymorphs of pharmaceuticals to hydrates and anhydrides.

Gemini での翻訳

Humidity is also considered **crucial** in various fields, such as managing low-humidity environments in battery manufacturing for electric vehicles as part of the move toward "decarbonization," and its effects on the quality and efficacy of pharmaceuticals due to the transition of crystal polymorphs into hydrates and anhydrides.

どちらの AI を選んだか

- **文の主語:** DeepL は "It is also important" と、形式主語 "It" を使っているが、これは文法的に正しいと考える。一方、Gemini の翻訳では、最初に "Humidity" を主語に置くことで、何が重要なのかをわかりやすくしている。
- **単語の選択:** DeepL の "important" (重要) に対し、Gemini は "**crucial**" (極めて重要) を選択しました。これは、論文全体の論調や、この技術が持つ意義の強さを鑑み、より強調する単語が適切と判断したためです。

目的

<日本語文>

既存の湿度計の精度が $\pm 5\%$ （高精度なものでも $\pm 2\%$ ）であるのに対し、テラヘルツ分光法を用いて水蒸気吸収スペクトルを測定・評価することで、1桁以上優れた精度の光湿度計を実現することを目的とする。

DeepL の翻訳

The objective is to realize an optical hygrometer with an accuracy more than one order of magnitude better than that of existing hygrometers, which have an accuracy of $\pm 5\%$ (or $\pm 2\%$ for high-precision models), by measuring and evaluating the water vapor absorption spectrum using terahertz spectroscopy.

Gemini の翻訳

To realize an optical humidity measurement system with an accuracy more than an order of magnitude better than existing hygrometers, which have an accuracy of $\pm 5\%$ (or $\pm 2\%$ for high-precision models), by using terahertz spectroscopy to measure and evaluate the water vapor absorption spectrum.

どちらの AI を選んだか

- **構文のスタイル**: DeepL の "The objective is to..." は、目的を説明する上で使用される正しい文法と調べた中では書いてあった。一方で、Gemini が選択した不定詞で始める "To realize..." という構文は、私が読んできた学術論文でよく使用されていたので、こっちの方がいいのではないかと思った。
- **用語の選択**: DeepL は「光湿度計」を "optical hygrometer" と訳しており、Gemini は "optical humidity measurement system" としている。ここは、専門的な用語になると思うので、短く意味の伝わりやすい DeepL を採用した。

考察

<日本語文>

測定点数の湿度ポイントを増やすことで回帰直線の精度が上がり、誤差が少なくなると考えられる。

DeepL の翻訳

It is thought that increasing the number of humidity points in the measurement will improve the accuracy of the regression line and reduce the error.

Gemini の翻訳

It is **believed** that increasing the number of humidity points for measurement **could** improve the accuracy of the regression line, thereby reducing the error.

どちらの AI を選んだか

- 「考えられる」という表現に対し、DeepL は "It is thought that" を使用しているが、Gemini は "**It is believed that**" を選んでいる。直感的に考えられるという意味なので、DeepL を選択した。
- DeepL の "will improve" は「改善するだろう」という未来の文法という感じがする。Gemini が用いた "**could improve**" は「改善する可能性がある」という表現になるため、Gemini 2.5 Pro を選択した。